

USC DEN Cover Sheet and Proctor Instructions

92.5

EE 558 Midterm

Student: Binh Phan	# of Pages: 17
Proctor: Lisa Madieros	Date: November 13, 2018

EXAM RULES

TIME LIMIT: 1 hour 20 minutes

RULES:

- 1) Exam is **CLOSED BOOKS/NOTES**.
- 2) Calculators are **NOT** allowed.
- 3) Do **NOT** write on the test pages. Use separate blank paper.

TOTAL EXAM PAGES: 4 pages (if student needs extra pages please provide it for them – thank you).

When the exam is complete please follow these steps to return the exam:

1. Please make a copy for your records.
2. Please **do not give** the exam or a copy to the student to keep.
3. Then use this cover sheet and e-mail to: examscans@den.usc.edu

PLEASE E-MAIL THE EXAM BACK ON THE SAME DAY AS THE TEST DATE. However, be sure that you receive confirmation from us that we have received the scanned version before mailing – thank you!

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EE 558: Midterm Exam

(Fall 2018)

Student Name : Binh Phan

Student ID # : 6530596568

Nov. 13th, 2018

Q. #.	Points
1	/ 1
2	/ 1
3	/ 5
4	/ 8
5	/ 12
6	/ 10
7	/ 10
8	/ 6
9	/ 8
10	/ 6
11	/ 12
12	/ 6
13	/ 7
14	/ 8
Total	/ 100

Exam is CLOSED BOOKS/NOTES.

Calculators are NOT allowed.

Do NOT write on the test pages. Use separate blank paper.

1. What does E. stand for in Alan E. Willner? (1 point)

2. Which company invented the low-loss fiber? (1 point) *Elvis*
Bell Labs

3. What do these acronyms stand for? (5 points)

dispersion compensating fiber

DCF - dispersion compensating fiber
WDM - wavelength division multiplexing
EDFA - erbium-doped fiber amplifier
DFB - dispersion fiber barrier
SPM - self-phase modulation
BER - bit error rate
SNR - signal-to-noise ratio
PMD - polarization mode dispersion
QAM - quadrature-amplitude modulation
NRZ - non-return to zero

4. Consider light traveling in medium 1 ($n_1 = 1.25$) is incident on an interface. The incident light makes an incident angle of 30 degrees and refracts into medium 2 ($n_2 = 1.2$). Determine the refracted and critical angles. What happens if the light is traveling from medium 2 to medium 1 with the same incident angle? (Hint: $\arcsin(25/48) \approx 31.4^\circ$, $\arcsin(3/4) \approx 48.6^\circ$, $\arcsin(4/9) \approx 28.7^\circ$) (8 points)

5. For data transmission, the bit rate is 20 Gb/s. Please use following parameters to design the system with exact chromatic dispersion compensation (i.e. the total dispersion at the end of link is zero). The fiber link is 300-km long, fiber dispersion at 1550 nm wavelength of SMF and DCF are 17 ps/nm/km and -85 ps/nm/km, respectively. The maximum transmission distance for SMF without compensation is 50 km. Draw the link (system) dispersion map (dispersion accumulation vs. distance) with corresponding fiber length, i.e. show the length of each kind of fiber along the link and the corresponding accumulated dispersion. (12 points)

6. (a) Describe chromatic dispersion and intermodal dispersion in fiber. (4 points)
(b) When the bit rate is doubled, how much worse is the effect of chromatic dispersion? (2 points)
(c) How does the number of spatial modes depend on the core size and the wavelength? (4 points)

7. (a) Describe briefly stimulated and spontaneous emission in an EDFA. (4 points)
(b) How do these two phenomena affect the SNR? (4 points)
(c) What is the function of an optical filter that follows the optical amplifier? (2 points)

8. Describe the following in one or two lines:
(a) Self-phase modulation (2 points)
(b) Cross-phase modulation (2 points)
(c) What happens if the fiber dispersion along the link is 0 ps/nm/km? (2 points)

9. In an optical fiber link, the EDFA bandwidth is about 28 nm (from 1532 nm to 1560 nm). How many 40 Gb/s channels can fit into this range for a WDM system? Suppose each 40 Gb/s

channel takes 40 GHz of bandwidth, and the separation between two adjacent channels should be no less than 10 GHz. (8 points)

10. (a) Name two different types of noise that are generated in a photodetector. (2 points)
(b) Describe the main characteristics of these noises. (4 points)
11. (a) Briefly describe direct detection and heterodyne detection, and give the main advantage and disadvantage of each. (4 points)
(b) Draw the diagram of heterodyne detection and derive the expression for the received heterodyne signal. (8 points)
12. Draw the eye diagram for NRZ signal, and show how it is generated. (6 points)
13. In an optical digital system, additive noise has the following probability density function: $0.5 \cdot \exp(-|V-V_m|)$. If the "zero" pulse's V_m corresponds to 2 mV and the "one" pulse's V_m corresponds to 14 mV at the receiver, what will the bit error rate (BER) be for the threshold level of 8 mV? (Your answer could be written in the form of $\exp(x)$) (7 points)
14. (a) Draw the data constellation for QPSK, 16-QAM and OOK. (3 points)
(b) For QPSK, how would it look if we had a filter at the receiver that was 10 times the signal bandwidth. (2 points)
(c) What is the spectral efficiency for each of these three data formats? (3 points)

Good Luck!

Useful Physical Constants and Formulas:

Planck's constant $h = 6.3 \times 10^{-34}$ joules*second

Speed of light $c = 3 \times 10^8$ m/s

Electron charge $e = 1.6 \times 10^{-19}$ coulombs

$$\# \text{ dB} = 10 \log(P_{\text{in}}/P_{\text{out}})$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B, \sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B, \cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)], \cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)], \sin A \sin B = -\frac{1}{2} [\cos(A + B) - \cos(A - B)]$$

$$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right), \sin A - \sin B = 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$$\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right), \cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

Question..... 1, 2, 3

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1. Elvis

2. Bell Labs

2. DFB - dispersion fiber barrier - 0.5

DCF - dispersion compensating fiber

SNR - signal-to-noise ratio

BER - bit error rate

WDM - wavelength division multiplexing

EDFA - erbium-doped fiber amplifier

QAM - quadrature amplitude modulation

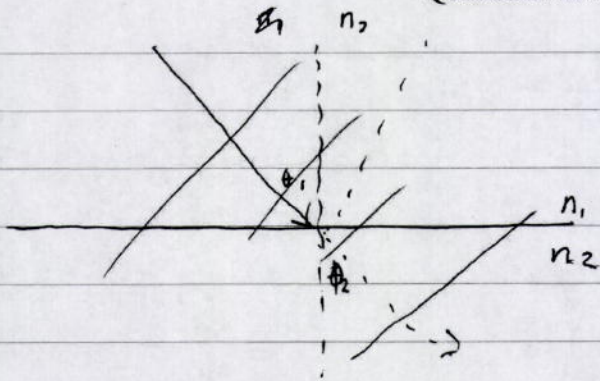
NRZ - non-return-to-zero

SPM - self-phase modulation

PMD - polarization mode dispersion.

Question.....4.....

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$$\theta = 30^\circ$$

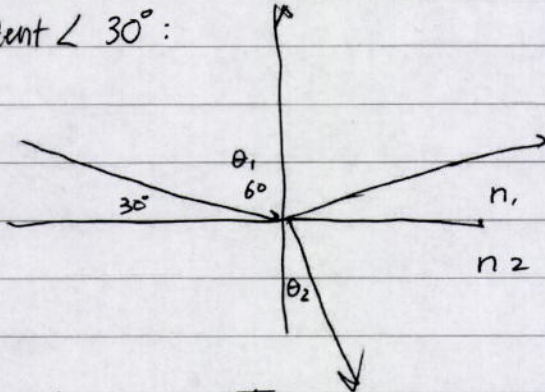
$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$1.25 \sin \theta_1 = 1.2 \sin \theta_2$$

$$\theta_2 = \sin^{-1} \left(\frac{1.25}{1.2} \sin \theta_1 \right)$$

$$\theta_2 = \arcsin \left(\left(\frac{1.25}{1.2} \right) \sin 30^\circ \right)$$

incident $\angle 30^\circ$:



$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\theta_2 = \sin^{-1} \left(\frac{n_1}{n_2} \sin \theta_1 \right)$$

$$= \sin^{-1} \left(\frac{1.25}{1.2} \left(\frac{1}{2} \right) \right)$$

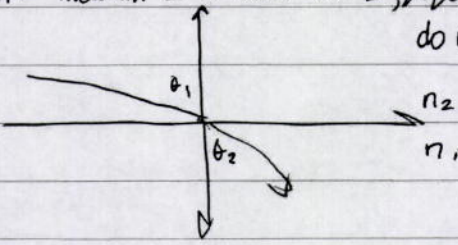
$$= \sin^{-1} \left(\frac{1.25}{2.4} \right)$$

$$= \sin^{-1} \left(\frac{25}{48} \right) = 31.4^\circ$$

when $\theta_2 = \frac{\pi}{2}$

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) = \arcsin \left(\frac{1.2}{1.25} \right)$$

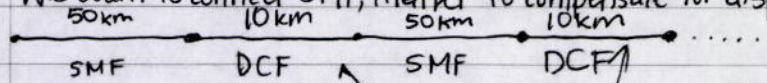
From medium 2 to medium 1, ~~we get~~ refracted angle changes, do not have a critical angle.



$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Question.....5

We want to connect SMF, then DCF to compensate for dispersion, like this:



total dispersion for 50km of SMF: $50 \times 17 \text{ ps/nmkm}$

total ~~amb~~ length needed for ~~10~~ DCF to compensate:

$$\frac{50 \times 17}{+ 85} = 10 \text{ km}$$

$$\begin{array}{r} 60 \\ \times 17 \\ \hline 350 \\ 500 \\ \hline 850 \\ \downarrow \\ 85 \end{array}$$

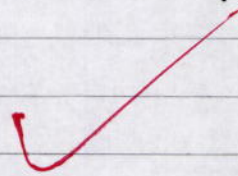
Dispersion map:



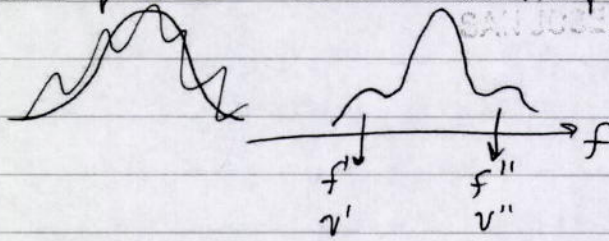
For 300km:

of ^{50km} SMF needed: 5

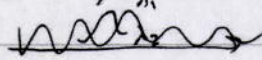
^{10km} DCF needed: 5



a. Chromatic dispersion is when you have a signal, some parts of the signal are at different frequencies $\rightarrow$ travel at different speeds.



b. ~~intermodal dispersion - different channels affect each other~~



$\rightarrow$ get "phase matching"

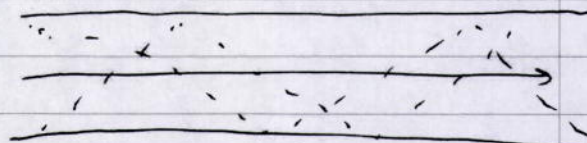
b. bit rate $\times 2$

CD effects $\times 4$

c. core size $\downarrow$, # of spatial modes $\downarrow$
wavelength $\downarrow$, # of spatial modes $\downarrow$

- |

intermodal dispersion:



signals travel at different angles,
reach the end of fiber at a
different time.

a. stimulated emission: light at the ~~proper~~ λ you want,
Good

spontaneous emission: random noise, Bad.

b. stimulated emission is your signal
sp. em. noise

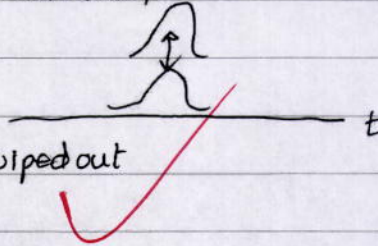
c. restrict BW to only your signal, reduce noise.

a. self-phase modulation

your signal interacts with itself + noise



b. XPM - ~~at~~ different λ channels affect your channel



c. you get phase matching, your signal is wiped out from nonlinearities.

28nm

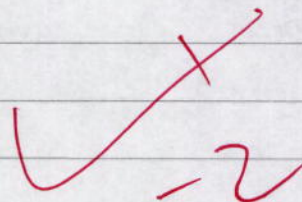
40Gb/s channel $\rightarrow$ ~~11~~ ~~11~~ ~~11~~ ~~11~~

each channel occupies $\lambda = \frac{1}{40 \text{eq}} = 4 \times 10^{-10} = 0.4 \text{nm} + 0.1 \text{nm spacing} = \underline{0.5 \text{nm/channel}}$

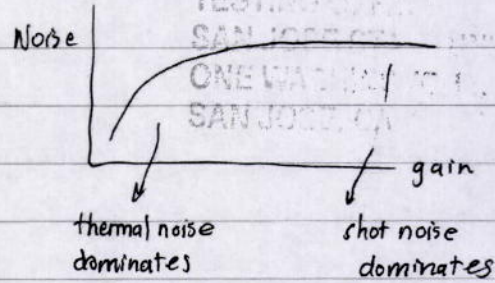
You can have ~~28~~ ~~channels~~

$$28 \times \cancel{2} \frac{\text{channel}}{\text{nm}} = \boxed{56 \text{ channels}}$$

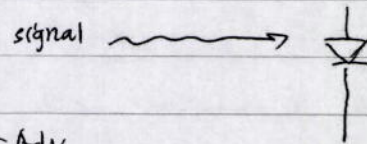
$$56 \frac{\text{GHz}}{\text{nm}} = \cancel{11} \frac{\text{GHz}}{\text{nm}} = \cancel{11} \frac{40 \text{GHz}}{\text{nm}} = \cancel{11} \frac{40 \text{GHz}}{30 \text{nm}}$$



- a. thermal noise, ~~shot~~ noise
- b. thermal noise depends on temp.
shot noise depends on gain(power)



a. direct detection -



signal goes to photodetector.

Adv. No pump needed ← Adv.

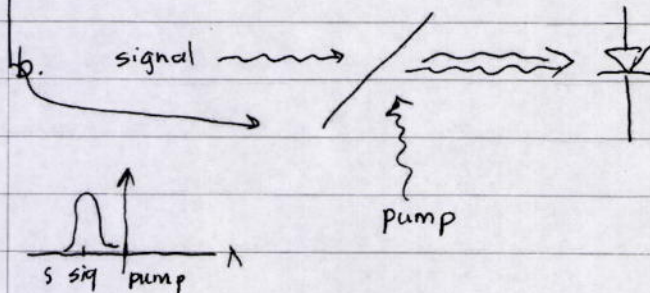
Disadv. Does not capture history of signal (phase) ← Disadv.

b. heterodyne detection

signal + pump goes to photodetector

Adv. → captures history of signal (phase)

Disadv. → needs pump signal



Two inputs: $E_{sig} = E \cos(2\pi f_{sig} t + \phi_{sig})$

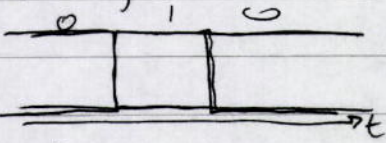
$E_p = E_p \cos(2\pi f_p t + \phi_p)$

terms cancel: terms cancel, output looks like:

$E_{output} = E(t) E_p \cos(2\pi(f_{sig} + f_p)t + (\phi_{sig} - \phi_p))$

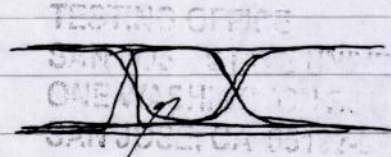
depends on ϕ of two signals!

NRZ signal:



all possible
combinations

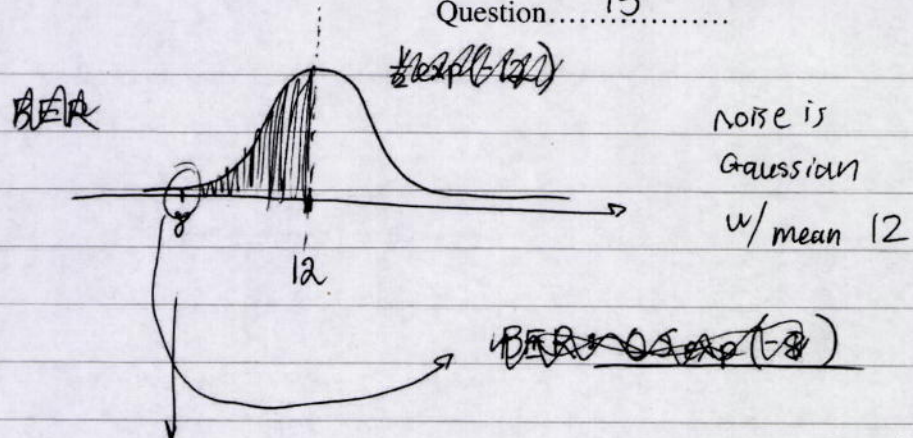
generated after going through
LPF filters, don't
get perfect
signal



eye diagram

eye: a qualitative measure of
signal quality

Question.....13.....



If threshold V is 8mV, may get error between 8 and 12mV!

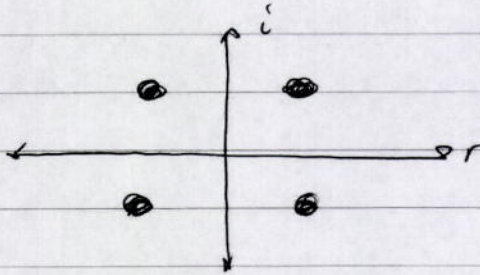
$$BER = \int_8^{12} 0.5 \exp(-|V-12|) dV$$

✓ -2

Question.....14.....

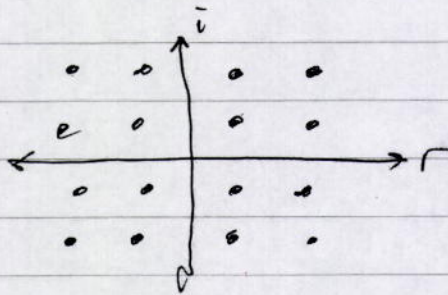
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a. QPSK

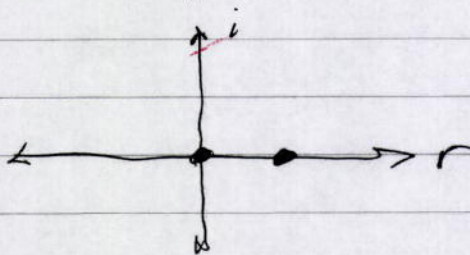


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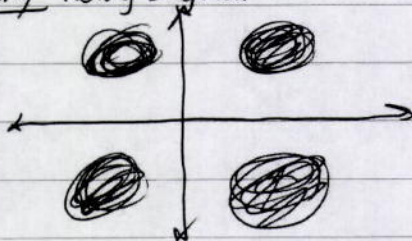
b. 16-QAM



c. OOK



b. we'd have a very noisy signal :



c. QPSK $\rightarrow 2$ bits/symbol time

16-QAM $\rightarrow 4$ bits/symbol time

OOK $\rightarrow 1$ bit/symbol time.

